LESSON 5.1

Determine the Area of Parallelograms and Rhombuses

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a parallelogram using base × height.

WHY THIS MATTERS Your architecture firm is designing a parallelogram-shaped patio for a client's backyard. The client needs to know how many square feet of pavers to order. The patio has a base of 14 feet and a height of 9 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Slanted side *Spanish: Lado inclinado*	A side that leans instead of going straight up and down. It is not the perpendicular height.	In a leaning parallelogram, use the dashed 90° height—not the slanted side—in A = b × h.
Parallelogram *Spanish: Paralelogramo*	A four-sided shape with two pairs of parallel sides.	opposite sides never meet
Parallel *Spanish: Paralelas*	Two lines or sides that stay the same distance apart and never meet.	The top and bottom of a parallelogram never get closer together — matching arrowheads (›) mark one parallel pair, double arrowheads (››) mark the other.
Base *Spanish: Base*	Any side of a parallelogram can be the base. Once you pick a base, the height must be measured perpendicular (at a 90° angle) to it.	b in A = b × h
Height *Spanish: Altura*	The straight-up distance from the base to the top.	h in A = b × h
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	height ⊥ base

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Base 9 in., height 4 in., slanted side 5 in. Find the area.

1. Base = 9 in., height = 4 in.
2. Skip the 5 in. slanted side.
3. A = 9 × 4 = 36

Answer: 36 in²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller parallelogram: base = 5 cm, height = 4 cm.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base is _____ and the perpendicular height is _____ .
- I know that measure is the height because _____ .
- Area = _____ × _____ = _____ square units.
- The slanted side is not used because _____ .

Word bank parallelogram · base · height · perpendicular · square corner · area · formula · multiply · square units · slanted side · rectangle · rearrange

WATCH OUT!

A common mistake

The most common mistake in Area of Parallelograms is multiplying the base by the slanted side instead of the perpendicular height. For a parallelogram with a base of 16 ft, a perpendicular height of 9 ft, and a slanted side of 11 ft, students compute 16 × 11 = 176 square feet instead of the correct 16 × 9 = 144 square feet. The height must be the straight-up (perpendicular) distance between the base and its opposite side, not the length of the slanted side, even though the slanted side is often the number that 'looks' like it belongs in the formula.

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

What is the area of a parallelogram with base 10 cm and height 7 cm?

(A) 17 sq cm
(B) 34 sq cm
(C) 70 sq cm
(D) 70 cm

STEPS TO TRY
1. You know both the base (10 cm) and the height (7 cm) — this is a direct use of the parallelogram area formula.
2. Area of a parallelogram is A = base × height. Adding the two numbers gives perimeter-style thinking, not area.
3. Set up A = 10 × 7 and multiply. Label your result in square centimeters.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

A parallelogram patio on the firm's blueprint has an area of 54 sq ft and a base of 9 ft. What is the height?

(A) 6 ft
(B) 27 ft
(C) 45 ft
(D) 63 ft

STEPS TO TRY

1. This time you're given the area (54 sq ft) and the base (9 ft) — the height is the missing piece.
2. Start from A = b × h. To undo a multiplication, divide: h = A ÷ b.
3. Write 54 = 9 × h, then divide 54 by 9 to find the height in feet.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

What is the area of a parallelogram with base 9 cm and height 5 cm?

(A) 14 sq cm
(B) 22.5 sq cm
(C) 45 sq cm
(D) 90 sq cm

STEPS TO TRY

1. Spot the base (9 cm) and the height (5 cm). Which operation does the area formula use on them?
2. Use A = base × height. If your number looks like 9 + 5, or like half the product, rethink the operation.
3. Compute 9 × 5 and give the answer in square centimeters.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

Which measurement is the height of a parallelogram?

(A) The length of the slanted side

(B) The perimeter divided by 4

(C) The perpendicular distance between the base and the opposite side

(D) The longest side

STEPS TO TRY

1. Picture a leaning parallelogram. The height is not a side you can trace — it's a distance measured straight across.
2. Height must make a right angle (90°) with the base. Test each choice: is this perpendicular to the base?
3. Draw a base, then a straight up-and-down dashed line to the opposite side. That dashed line is what 'height' means here.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

If you cut a triangle from one side of the parallelogram and move it to the other side, what shape do you get? How does this help you find the area?

SENTENCE STARTER The area of the parallelogram equals base x height = ___ x ___ = ___ square feet, which is the same as a ___ with the same base and height.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help